

Exploiting the Source-Asymmetry Confidence Gap for Generalizable AI-Generated Image Detection

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Abstract

Existing AI-generated image (AIGI) detection methods are trained on images synthesized by only a few generators, resulting in poor generalization to the ever-evolving domain of unseen generative models. In response to this limitation, recent work leverages large Vision Foundation Models (VFM) to obtain more robust representations, yet typically relies on computationally expensive strategies, e.g., introducing additional large-scale synthetic datasets or employing complex training pipelines. In this paper, we devise a lightweight VFM-based strategy for generalized AIGI detection. Our key insight is a **source asymmetry** phenomenon: during training, models are exposed to diverse real data, while the fake images are synthesized by only a few generators. We verify that this class-diversity asymmetry causes a fine-tuned VFM to learn a more generalizable representation for the real class. This mechanism yields our key finding: the **Source-Asymmetry Confidence Gap (SACG)**, a consistent and exploitable separation in predictive confidence where unseen real images yield systematically higher confidence than unseen fakes. Leveraging this insight, we introduce a lightweight, post-hoc strategy that employs a specialized confidence metric (GEN score) to amplify this gap and applies a conditional decision rule calibrated only on the real images for stable detection. On the Community Forensics benchmark (21 distinct generative models), our method achieves state-of-the-art results, surpassing the next-best baseline by **+6.82%** in accuracy.

1. Introduction

The rapid evolution and proliferation of generative models, from open-source diffusion systems [37] to advanced proprietary models like DALL-E 3 [31], pose an escalating challenge to digital forensics. While existing AI-generated

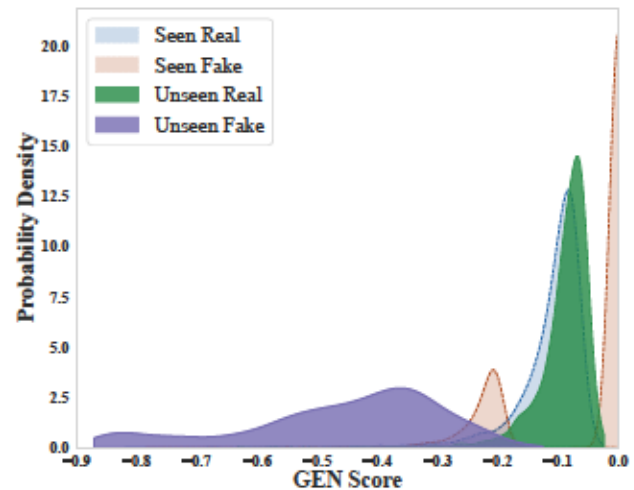


Figure 1. **The Source-Asymmetry Confidence Gap (SACG)**. Probability density of the GEN score [21] (see Sec. 3.3.1), where scores closer to 0 indicate higher confidence. Most seen samples (blue and orange) and unseen real (green) samples yield high-confidence scores, while unseen fake (purple) samples form a distinct, lower-confidence distribution. Our method (see Sec. 3) exploits this gap by calibrating a decision threshold using only the seen real images, enabling robust generalization to unseen generators. The visualization is obtained from a fine-tuned CLIP-ViT.

image (AIGI) detectors can achieve high accuracy on generators seen during training, their performance often degrades substantially when applied to images from unseen generators [43]. This *generalization gap* is widely recognized as the critical bottleneck hindering the deployment of robust, real-world detectors.

Recent state-of-the-art approaches demonstrate that leveraging large Vision Foundation Models (VFMs), such as CLIP [35] and DINOv2 [32], provides robust feature representations for AIGI detection [18, 45, 46]. However, ex-

isting strategies to enhance generalization via VFMs often resort to computationally expensive paradigms. One line of work, for instance, requires introducing extensive, large-scale datasets of generated images to expose the model to diverse artifacts [33]. Alternatively, other approaches devise complex training workloads. These include designing intricate network architectures [18, 46] or formulating sophisticated fine-tuning objectives, such as subspace decomposition [45] or contrastive frameworks with hard-negative mining [5]. This reliance on computationally intensive interventions leaves a critical research gap: the development of a lightweight and efficient strategy that can harness the generalization power of VFMs without demanding massive auxiliary data or complex training schemes.

In this paper, to devise a lightweight VFM-based strategy for generalized AIGI detection, we shift our perspective from designing complex training schemes to exploiting a fundamental yet insufficiently leveraged property of the training data: the *source asymmetry* phenomenon. This refers to the training condition where a detector has access to diverse, semantically-rich real data but only a limited set of generated fake data from a few known sources. We first verify that this class diversity asymmetry leads a fine-tuned VFM to learn a more generalizable representation for the real class. Supporting evidence shows that the real class feature space retains a broad, high-dimensional structure, whereas the fake class representation collapses into a compact, low-dimensional subspace (see Fig. 2 and Tab. 1). Consequently, the detector’s predictive confidence on unseen real images is systematically higher than on unseen generated fakes, as visualized in Fig. 1. This core observation allows us to identify the **Source-Asymmetry Confidence Gap (SACG)**—a consistent and exploitable separation in predictive confidence between real and generated image distributions.

To this end, we leverage SACG through a lightweight, post-hoc strategy. First, we employ a specialized confidence metric, the GEN score [21], to transform the detector’s output probabilities into a new confidence score specifically designed to amplify this separation. These resulting confidence scores are used to effectively isolate high-confidence samples (scores near 0), for which the detector’s original prediction is trusted. Then, for the remaining samples, which fall into the lower-confidence spread (visualized in Fig. 1), we apply a conditional decision rule. Crucially, the threshold for this rule is calibrated only on the distribution of the seen real images. This simple, real-only calibration enables our method to stably separate the *hard* real samples from the unseen fake samples that predominantly populate this low-confidence region. The entire procedure is implemented as a post-hoc strategy, requiring no architectural changes and incurring zero additional training overhead.

We validate our approach on two comprehensive bench-

marks. On the GenImage benchmark [47] (7 unseen generators), our method establishes a new state-of-the-art, surpassing the next-best baseline by +4.09% in accuracy. We demonstrate even stronger performance on the larger and more challenging Community Forensics benchmark [33] (21 distinct generative models), where we again establish a new state-of-the-art, surpassing the next-best baseline by +6.82% in accuracy. This SOTA performance across diverse benchmarks is achieved without complex training workloads or additional auxiliary data, highlighting the efficacy of our lightweight strategy.

Our primary contributions are:

- We systematically investigate the consequences of source asymmetry, verifying that this data imbalance causes a fine-tuned VFM to learn an asymmetric feature-space representation.
- We identify and validate that this feature-space asymmetry produces the Source-Asymmetry Confidence Gap (SACG) at the classifier’s output—a novel and reliable signal for generalization.
- We propose a lightweight, post-hoc detection strategy that efficiently exploits SACG. The method requires calibration only on the seen real images, enabling it to stably generalize to unseen generators without complex training schemes or architectural overhead.

2. Related Work

Research in AIGI detection has primarily followed several dominant paradigms. Among them, approaches leveraging large-scale Vision Foundation Models (VFMs) [32, 35] have recently shown significant promise.

Forgery Artifact-Centric Detectors. One paradigm assumes that generated images contain generator-related forgery artifacts. Foundational work suggested that CNN-based generators shared common flaws [41]. More recent methods use VFMs as feature extractors to hunt for these artifacts, combining low-level frequency analysis with high-level semantic features [44], or pixel artifacts with semantic inconsistencies [6]. Rather than pursuing these artifact-based signals, which are inherently reactive to generator evolution, our work investigates a more fundamental property of the training data itself (*source asymmetry*) to find a stable generalization signal.

Advanced Training and Architectural Strategies. To improve generalization, another line of work proposes more complex training schemes or architectures. Data-centric methods employ contrastive frameworks with re-constructed hard-negative samples [5], create semantically-identical real/fake pairs to mitigate content bias [15] or introduce additional large-scale generated images [33]. Another sub-trend avoids the cost of full fine-tuning by adapting pre-trained backbones in parameter-efficient ways. These approaches range from using completely frozen fea-

tures with simple nearest-neighbor searches [29], to training lightweight networks on representations extracted from intermediate encoder blocks [18], to more sophisticated subspace decomposition methods [45]. While we also leverage a VFM-based detector, we introduce no architectural complexity or expensive training objectives.

Modeling the real Image Distribution. The third paradigm focuses on modeling the distribution of real images and treating fakes as anomalies. Some methods model low-level statistical properties [7, 17, 20]. Other training-free approaches use reconstruction error from pre-trained diffusion models [42] or their internal autoencoders [36] as a signal of authenticity. Unlike these approaches, which treat detection as a one-class or out-of-distribution (OOD) problem, our method trains a binary classifier and exploits a post-hoc signal (SACG) that emerges from the asymmetric generalization between the two learned classes.

3. Methodology

This section provides a detailed analysis of the *source asymmetry* phenomenon and introduces our method to exploit its consequences. We first investigate how this asymmetry leads to the observed **Source-Asymmetry Confidence Gap (SACG)**. We then present our lightweight, post-hoc strategy that leverages this gap for stable, generalized detection.

3.1. Problem Setup

Let the training data $\mathcal{D}_{\text{train}} = \mathcal{R}_{\text{seen}} \cup \mathcal{F}_{\text{seen}}$, where $\mathcal{R}_{\text{seen}}$ is a diverse set of real images and $\mathcal{F}_{\text{seen}}$ is a set of fake images sourced from a limited collection of seen generators, $\mathcal{G}_{\text{seen}} = \{G_1, \dots, G_n\}$. We train a binary AIGI detector f , which comprises a Vision Foundation Model (VFM) backbone (e.g., CLIP-ViT [12, 35]) and an MLP head, using a standard binary cross-entropy (BCE) loss. We assign labels $y = 0$ for the real class and $y = 1$ for the fake class. Given an input x , the detector f outputs a sigmoid probability $p = f(x) \in [0, 1]$, representing the predicted probability $P(y = 1|x)$.

The objective is to ensure that f , trained only on $\mathcal{D}_{\text{train}}$, generalizes stably to a test set $\mathcal{D}_{\text{test}}$. This set contains real images and fake images from a disjoint set of unseen generators, $\mathcal{G}_{\text{unseen}} = \{G_{n+1}, \dots, G_m\}$, where $\mathcal{G}_{\text{seen}} \cap \mathcal{G}_{\text{unseen}} = \emptyset$.

3.2. Source-Asymmetry Confidence Gap

This section analyzes the mechanism underlying the Source-Asymmetry Confidence Gap (SACG).

We identify SACG as a direct consequence of the source asymmetry (see Sec. 3.1) inherent in the training data. Specifically, the fine-tuning process, when exposed to the highly diverse real set $\mathcal{R}_{\text{seen}}$, learns a broad and generalizable representation for the real class. Conversely, the limited, low-diversity fake set $\mathcal{F}_{\text{seen}}$ forces the model to learn a collapsed, source-specific representation for seen fakes.

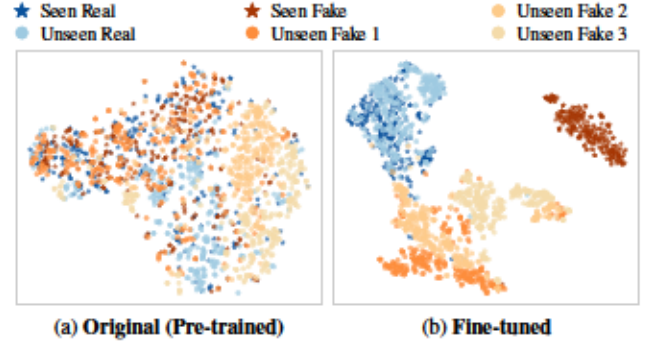


Figure 2. **Asymmetric Feature Separation.** t-SNE [25] visualization of embeddings from (a) the original and (b) the fine-tuned CLIP-ViT backbone. (a) The original embeddings show no class separability. (b) Fine-tuning induces a *structural asymmetry*: the real class (seen and unseen) forms a single, broad cluster, demonstrating a generalizable representation. In contrast, the seen fake class collapses into a tight, distinct group, separate from unseen fakes, indicating a source-specific representation.

ited, low-diversity fake set $\mathcal{F}_{\text{seen}}$ forces the model to learn a collapsed, source-specific representation for seen fakes.

Asymmetric Feature Separation. We visualize the VFM backbone embeddings using t-SNE [25] in Fig. 2. As shown in Fig. 2a, the pre-trained features show no clear class separability. In contrast, the fine-tuned features in Fig. 2b reveal a distinct **structural asymmetry**. The *seen real* and *unseen real* samples form a cohesive and broad cluster, demonstrating that the model has learned a generalizable representation for the real class. Conversely, the *seen fake* samples collapse into a tight, isolated cluster, while *unseen fake* samples map to separate regions. This visually confirms that the learned fake representation is non-generalizable and highly specific to the training sources.

Representation Complexity Collapse. We quantify this structural asymmetry by measuring the effective dimensionality of the class representations. As shown in Tab. 1, we compute the number of principal components (PCs) required to explain 95% of the variance for the real and fake class embeddings. The pre-trained features exhibit high, comparable dimensionality (211 vs. 233 PCs). After fine-tuning, a significant **asymmetric collapse** occurs: the fake class dimensionality plummets to just 15 PCs, while the real class retains a significantly higher dimensionality of 71 PCs. This quantitatively validates that fine-tuning on a source-asymmetric dataset yields a generalizable, high-dimensional representation for the real class and a collapsed, low-dimensional representation for the fake class.

3.3. Proposed Method

Our method is a lightweight, post-hoc strategy designed to efficiently exploit SACG. It consists of three key components, detailed below: a specialized confidence metric to

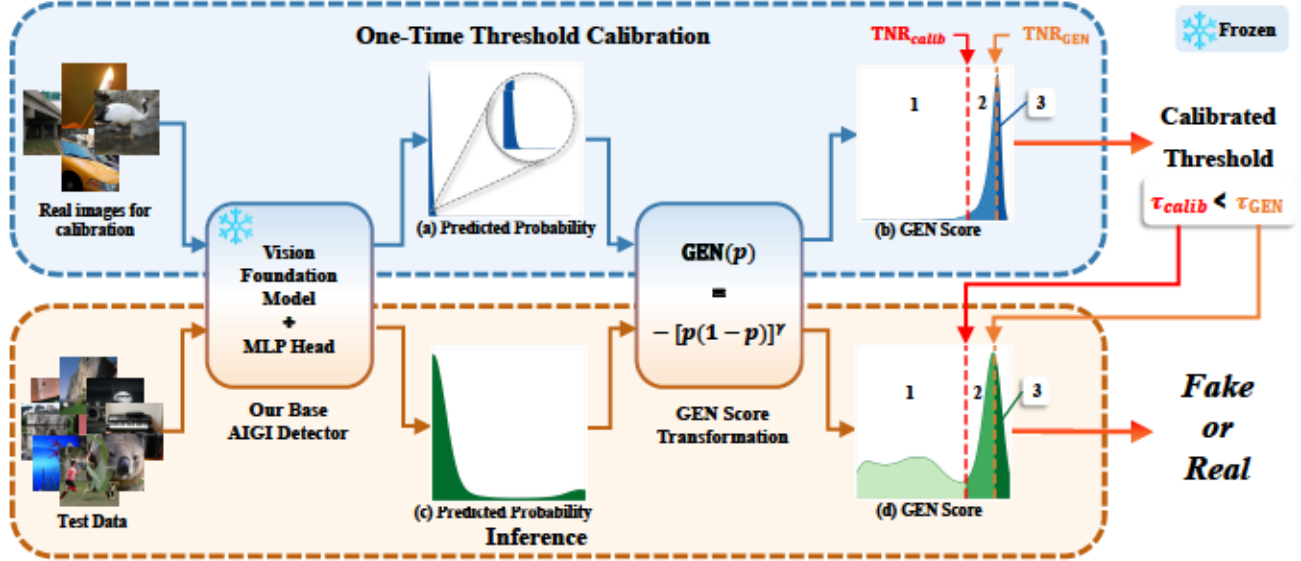


Figure 3. **Our Post-Hoc Decision Pipeline.** The pipeline operates in two phases. **(Top) One-Time Threshold Calibration:** A small set of seen real images are processed to get Predicted Probabilities (a) and then GEN scores (b). From this real-only distribution, we pre-calibrate two thresholds, τ_{calib} and τ_{GEN} , based on user-defined True Negative Rate (TNR) constraints, where $0\% < TNR_{GEN} < TNR_{calib} < 100\%$ (see Sec. 3.3.3). **(Bottom) Inference:** For a test image, the Base AIGI Detector’s output (c) is transformed into a $GEN(p)$ score (d). Our conditional strategy (Eq. (2)) then uses τ_{GEN} to trust the detector’s original prediction for high-confidence samples (Region 3). For the remaining low-confidence samples, τ_{calib} is used to make the final decision: samples in Region 1 ($GEN(p) < \tau_{calib}$) are predicted as **Fake**, while those in Region 2 are predicted as **Real**.

Class	Original (Pre-trained)	Fine-tuned
real	211	71
fake	233	15

Table 1. **Representation Complexity Collapse.** Number of principal components required to explain 95% variance for each class’s embeddings. Fine-tuning induces an asymmetric collapse in effective dimensionality: the fake class representation collapses, while the real class retains a significantly higher dimensionality. This provides quantitative validation for the asymmetric generalization shown in Fig. 2.

amplify the gap (Sec. 3.3.1), a conditional decision strategy to apply this signal (Sec. 3.3.2), and a simple, real-only calibration for that strategy (Sec. 3.3.3).

3.3.1. Metric to Amplify the Confidence Gap

A standard AIGI detector’s final sigmoid activation compresses outputs p toward the high-confidence boundaries (0 or 1). This compression renders metrics like Shannon entropy sub-optimal, as they are insufficiently sensitive to the small yet critical deviations within this region (e.g., distinguishing $p = 0.01$ from $p = 0.001$). This is a key challenge, as the observed SACG (see Fig. 1) manifests precisely as a separation within these subtle, high-confidence deviations. Therefore, to effectively exploit this gap, we require a met-

ric specifically designed to amplify these deviations near the 0 and 1 boundaries.

We select the GEN score [21] for this purpose. Given the detector’s output probability p , the score is computed as:

$$GEN(p) = -[p(1-p)]^\gamma \quad (1)$$

The metric’s sensitivity is governed by the exponent γ . Following the original work [21], we set $\gamma = 0.1$. This small value makes the metric highly responsive to small changes near the $p \rightarrow 0$ and $p \rightarrow 1$ boundaries, effectively magnifying the signal in the high-confidence region. As visually demonstrated in Fig. 1 and further validated in our ablations (Sec. 4.3), this metric provides a much cleaner separation of the confidence distributions than standard entropy.

3.3.2. Conditional Decision Strategy

Our strategy is motivated by the observation from Fig. 1: the GEN score distribution reveals two distinct regions. First, there is a high-confidence regime where $GEN(p)$ scores cluster near 0. This cluster contains both *easy* real and *easy* fake samples, representing cases where the detector’s original probability p is already decisively close to 0 or 1. For these samples, the detector’s predictions are highly reliable. We therefore define a threshold, τ_{GEN} , to isolate this regime. Second, there is a lower-confidence regime (all samples with $GEN(p) \leq \tau_{GEN}$). This region contains the

hard cases. Critically, this is the precise region where the observed **SACG**—the separation between *hard* real samples and unseen fake samples—becomes the dominant, exploitable signal.

This two-regime structure motivates our conditional decision strategy, illustrated in Fig. 3:

$$\hat{y} = \begin{cases} \mathbb{I}(p > 0.5) & \text{if } \text{GEN}(p) > \tau_{\text{GEN}}, \\ \mathbb{I}(\text{GEN}(p) < \tau_{\text{calib}}) & \text{otherwise.} \end{cases} \quad (2)$$

This policy operates as follows:

- **If** $\text{GEN}(p) > \tau_{\text{GEN}}$: The sample is in the high-confidence regime. We trust the base detector’s original prediction, $\mathbb{I}(p > 0.5)$.
- **Otherwise**: The sample falls into the lower-confidence regime. Here, we apply the SACG-derived strategy, using a second threshold, τ_{calib} , to separate the unseen fakes from the hard real samples. A sample is classified as fake ($\hat{y} = 1$) only if its score falls below this threshold.

3.3.3. Real-Only Threshold Calibration

A key principle of our method is that both thresholds, τ_{GEN} and τ_{calib} , are calibrated post-hoc using only a small subset of the seen real images ($\mathcal{S}_{\text{real}} \subset \mathcal{R}_{\text{seen}}$). This real-only calibration ensures our method is independent of the seen fake generators $\mathcal{G}_{\text{seen}}$, which is critical for generalization. Both thresholds are derived using a single, unified mechanism: finding the $\text{GEN}(p)$ score that corresponds to a target True Negative Rate (TNR) on this real calibration set. Formally, given a target $\text{TNR}_{\text{target}} \in [0, 1]$, we find the corresponding threshold τ by solving for the quantile:

$$\frac{1}{|\mathcal{S}_{\text{real}}|} \sum_{x \in \mathcal{S}_{\text{real}}} \mathbb{I}(\text{GEN}(f(x)) \geq \tau) = \text{TNR}_{\text{target}} \quad (3)$$

This mechanism is applied to find both thresholds:

- τ_{calib} : We set this by choosing a high, user-defined operating point, such as $\text{TNR}_{\text{calib}} = 95\%$. This deterministically sets τ_{calib} to the value that correctly classifies 95% of the real calibration samples.
- τ_{GEN} : We set this by choosing a second, intermediate target, TNR_{GEN} , such that $0 < \text{TNR}_{\text{GEN}} < \text{TNR}_{\text{calib}}$. This effectively isolates the high-confidence peak from the lower-confidence region where the SACG is exploited. As we demonstrate in our ablation study (Sec. 4.3), our method is robust to a wide range of TNR_{GEN} values, making precise tuning unnecessary.

4. Experiments

4.1. Experimental Setup

Datasets and Benchmarks. The VFM-based detectors we use in our post-hoc strategy are trained on the GenImage

dataset [47]. The training set consists of 162k real images (ImageNet [10]) and 162k fake images from a single generator (SD V1.4 [37]).

For evaluation, we test generalization on two diverse benchmarks: 1) The **GenImage Benchmark** [47], containing fakes from 7 unseen generators (*e.g.*, BigGAN [4], ADM [11]), and 2) The **Community Forensics Benchmark** [33], a challenging benchmark with fakes from 21 distinct models (*e.g.*, DALL·E 3 [31], Imagen 3 [13]).

Implementation Details. We use CLIP-ViT [35] as the VFM backbone, with other backbones explored in Sec. 4.3. We train our base AIGI detector by partially fine-tuning only the latter half of the backbone, which is followed by a two-layer MLP head. We train for 10 epochs using AdamW [23] (batch size 128) with differential learning rates: 1×10^{-5} for the backbone and 1×10^{-3} for the MLP head. All other protocols follow Rine [18].

Calibration and Metrics. Our post-hoc strategy is applied at inference time. The GEN score exponent is $\gamma = 0.1$ [21]. As defined in Sec. 3.3.3, both thresholds are calibrated using a small subset of 128 seen real images ($\mathcal{S}_{\text{real}} \subset \mathcal{R}_{\text{seen}}$) via our unified TNR-based mechanism. We use $\text{TNR}_{\text{calib}} = 95\%$ and $\text{TNR}_{\text{GEN}} = 20\%$. We evaluate all methods using Accuracy (Acc).

Baselines. We compare against a comprehensive suite of SOTA detectors: CNNSpot [41], LNP [20], UnivFD [29], DIRE [42], NPR [38], DRCT [5], Rine [18], AIDE [44], VIBAIGCDetect [46], and Effort [45]. To ensure fairness, we used official codebases. Methods not originally trained on GenImage [18, 29, 41] were retrained by us. For methods already trained on GenImage [38, 44–46] or those with unique data requirements [5, 20, 42], we used their official, pre-trained models as per author instructions.

4.2. Main Results

We evaluate our lightweight, post-hoc strategy against ten SOTA baselines on two challenging benchmarks. We apply our method to two VFM backbones: a base CLIP-ViT-B/16 and a larger CLIP-ViT-L/14, to demonstrate its effectiveness and scalability. The full accuracy results are presented in Tab. 2 and Tab. 3.

Performance on the GenImage Benchmark. As shown in Tab. 2, our approach establishes a new state of the art. Our ViT-B/16 model achieves 95.37% average accuracy, and our ViT-L/14 model further improves to 96.10%. This decisively surpasses the next-best baseline, Effort [45] (92.01%), by +3.36% and +4.09%, respectively. This robust performance across all generator families (*e.g.*, BigGAN [4], ADM [11], Glide [28]) validates that exploiting the observed SACG is a highly effective strategy for generalization.

Performance on the Community Forensics Benchmark. On the more diverse and challenging Community Forensics

Table 2. **Generalization performance on the GenImage benchmark [47].** All our models were trained on the SD V1.4 subset of GenImage [47]. Accuracy (%) is reported. Baselines used models trained on the same data, or their official public models [5, 20, 42]. Our method with a ViT-B/16 backbone achieves **95.37%** average accuracy, already surpassing the best baseline (Effort [45], 92.01%). Our ViT-L/14 model extends this lead to **96.10%**. Best and second-best results are in **bold** and underlined.

Test Sets	CNNSpot [41]	LNP [20]	DIRE [42]	UnivFD [29]	NPR [38]	Rine [18]	DRCT [5]	AIDE [44]	VIBDetect [46]	Effort [45]	Ours ViT-B/16	Ours ViT-L/14
BigGAN [4]	51.85	63.51	51.76	56.01	81.84	78.62	81.67	66.83	55.57	92.28	<u>92.02</u>	89.20
ADM [11]	54.38	73.48	49.06	79.27	84.63	83.78	79.42	78.49	74.42	84.91	<u>96.02</u>	97.65
Glide [28]	67.18	72.86	51.68	88.53	90.31	89.91	89.18	91.78	73.36	92.58	<u>96.24</u>	98.43
VQDM [14]	53.62	63.74	50.89	90.20	87.04	99.18	90.03	80.24	89.21	93.40	96.23	<u>98.27</u>
SD V1.5 [37]	99.94	61.53	50.41	99.37	90.74	<u>99.81</u>	94.41	99.75	99.29	98.65	95.98	98.06
Midjourney V5 [26]	54.88	54.07	48.42	81.13	89.84	81.49	91.50	81.38	88.28	84.36	95.02	<u>92.80</u>
Wukong [27]	<u>99.85</u>	60.36	50.41	98.50	90.67	99.88	94.67	98.85	98.68	97.92	96.11	98.29
Avg.	68.82	64.22	50.38	84.71	87.87	90.38	88.70	85.33	82.69	92.01	<u>95.37</u>	96.10

benchmark (Tab. 3), our method’s superiority is even more pronounced. Our ViT-B/16 model achieves 97.49% average accuracy, surpassing the strongest baseline, Effort [45], by a significant margin of +6.11%. Our ViT-L/14 model extends this lead, achieving 98.20% accuracy. Notably, our performance remains exceptionally high against contemporary commercial models like DALL-E 3 [31] and Imagen 3 [13]. This confirms that the SACG is a stable signal for generalization, even against the most rapidly evolving, real-world generators.

Efficiency and Scalability of SACG Strategy. These results highlight a crucial observation regarding efficiency and scalability. Many SOTA baselines [5, 18, 29, 44–46] rely on computationally expensive training schemes and large ViT-L backbones to achieve their performance. In contrast, our lightweight, post-hoc strategy, when applied to a smaller ViT-B/16 backbone, already outperforms these heavier, more complex approaches. As seen on the Community Forensics benchmark, our ViT-B/16 (97.49% Acc) decisively surpasses the ViT-L-based Effort (91.38% Acc). Moreover, our simple, backbone-agnostic strategy scales effectively, with our ViT-L/14 model extending this lead to 98.20%. This confirms that exploiting the SACG is a more efficient and effective path to generalization than relying on complex, training-heavy designs.

4.3. Ablation Studies

We conduct a suite of ablation studies to validate our design choices and dissect the contribution of each component. Unless otherwise specified, all ablations use a fine-tuned CLIP-ViT-B/16 model.

Strategy Component and Metric Ablation. The **Base Detector**, using a $p > 0.5$ rule, yields a baseline accuracy of 70.35%. Simply applying our full strategy (**Ours**)

to this same detector boosts performance to 96.43%. This confirms our gain stems from effectively exploiting the observed SACG. This effectiveness hinges on two key design choices. First, the choice of metric is critical: replacing the GEN score with *Shannon entropy* causes a performance collapse to 13.98%. This validates our analysis in Sec. 3.3.1 that Shannon entropy is insensitive to the high-confidence signals. Second, our conditional, two-threshold structure is vital. A simpler variant, **Ours (w/o τ_{GEN})**, which applies the τ_{calib} rule globally, achieves 90.90% accuracy. While still effective, it underperforms our full strategy by 5.53%, confirming that isolating the *easy* high-confidence cases with τ_{GEN} is a more robust approach.

Hyperparameter Sensitivity Analysis. As summarized in Tab. 5, our method is highly robust to its key parameters. Performance is stable around our default $\text{TNR}_{\text{calib}}$ of 95% and is highly data-efficient, achieving optimal accuracy with only 128 seen real images for calibration. This optimal data size represents a balance: the performance drop at 32 images is due to a poor statistical approximation of the real distribution, leading to a biased threshold. Conversely, the slight drop at 512 images is because this larger sample introduces more *hard* real samples (the low-confidence tail). To maintain the 95% TNR target, τ_{calib} is forced to a lower confidence value, which inadvertently misclassifies more unseen fakes as real. Crucially, the TNR_{GEN} parameter, which separates the high-confidence regime, is exceptionally insensitive. As shown in the table, performance is identical across the entire tested range (1% to 60%). This confirms our claim in Sec. 3.3.3 that TNR_{GEN} does not require precise hyperparameter tuning and is robust by default.

Generality Across VFM Backbones. To confirm our strategy is not tied to a specific architecture, we apply it to base

Table 3. **Zero-shot generalization to the Community Forensics Benchmark [33].** We report Accuracy (%) on 21 diverse generators using the *exact same models* from the GenImage experiments (Tab. 2). Our ViT-B/16 model achieves **97.49%** *All Avg.* accuracy, surpassing the strongest baseline (Effort [45], 91.38%) by **+6.11%**. Our ViT-L/14 model extends this lead to **98.20%**. Our method is also SOTA on the *Commercial Avg.* (representing the **bottom 11** generators), achieving **97.60%** and **97.78%** for ViT-B and ViT-L, respectively. Best and second-best results are in **bold** and underlined.

Test Sets	CNNSpot [41]	LNP [20]	DIRE [42]	UnivFD [29]	NPR [38]	Rine [18]	DRCT [5]	AIDE [44]	VIBDetect [46]	Effort [45]	Ours ViT-B/16	Ours ViT-L/14
DF-GAN [39]	50.15	83.10	51.65	98.75	88.30	99.65	95.30	76.25	98.55	96.95	98.20	<u>99.20</u>
LCM-SDv1.5 [24]	63.70	53.10	51.40	96.70	92.20	98.40	93.10	93.85	95.30	95.90	96.40	<u>98.20</u>
LCM-SDXL [24]	50.00	42.45	51.60	97.15	90.65	<u>98.80</u>	93.40	72.50	97.30	94.15	98.00	99.35
LCM-SSD1B [24]	50.05	41.55	48.90	97.10	86.85	99.80	94.30	77.40	97.40	95.40	97.85	<u>99.05</u>
GALIP [40]	50.10	44.40	51.50	91.55	84.45	68.15	92.60	74.95	91.40	89.05	<u>97.80</u>	99.35
Stable Cascade [34]	69.10	70.15	51.30	99.15	92.75	99.75	94.15	94.90	<u>99.35</u>	97.10	97.90	99.25
Kvikontent [19]	<u>99.65</u>	63.65	51.40	99.45	93.10	100.00	95.60	99.20	98.95	96.60	97.95	99.45
Kandinsky V2.2 [2]	90.10	53.40	51.30	99.35	91.20	100.00	94.80	89.15	99.25	96.50	98.40	<u>99.55</u>
HDiT [8]	52.95	67.00	51.20	58.90	85.85	63.05	68.60	79.55	58.45	80.65	96.15	<u>95.70</u>
Deci Diff V2 [9]	64.30	52.25	51.25	83.65	92.65	91.20	93.55	92.85	93.30	<u>96.05</u>	95.10	97.55
Midjourney V5 [26]	53.16	53.84	50.05	90.29	84.55	92.62	78.22	80.61	82.29	94.13	<u>97.22</u>	98.22
Midjourney V6 [26]	53.93	45.77	50.10	95.17	80.29	<u>99.10</u>	90.62	85.02	95.22	89.94	98.05	99.30
DALL-E 2 [30]	51.95	47.85	49.50	92.85	91.15	<u>97.60</u>	91.05	75.80	92.20	88.20	97.10	98.70
DALL-E 3 [31]	98.00	49.85	49.95	99.40	92.45	100.00	94.10	99.20	98.70	96.25	98.00	<u>99.60</u>
Firefly 2 [1]	50.05	41.49	50.83	52.21	75.93	50.18	90.06	61.50	74.49	68.06	97.32	<u>92.45</u>
Firefly 3 [1]	50.10	54.72	48.31	56.34	77.85	51.26	88.89	64.91	84.37	71.18	98.10	<u>92.61</u>
Ideogram V1 [16]	77.60	48.35	49.95	95.80	91.20	<u>98.15</u>	85.50	87.70	92.05	96.60	97.00	99.00
Ideogram V2 [16]	79.35	48.10	49.80	94.20	90.75	97.40	79.55	84.20	90.90	96.10	<u>97.80</u>	99.00
Imagen 3 [13]	74.60	53.64	50.14	97.30	91.01	100.00	89.97	81.93	94.51	89.74	98.20	<u>99.48</u>
Flux-dev [3]	64.45	63.80	50.85	74.55	91.40	84.95	84.10	93.80	82.45	95.35	<u>97.60</u>	98.80
Flux-schnell [3]	57.50	59.80	50.00	66.35	91.00	76.85	80.40	88.25	74.95	95.10	<u>97.20</u>	98.45
Commercial Avg.	64.61	51.57	49.95	83.13	87.05	86.19	86.59	82.08	87.47	89.15	<u>97.60</u>	97.78
All Avg.	64.32	54.20	50.52	87.44	88.36	88.90	88.95	83.50	90.07	91.38	<u>97.49</u>	98.20

Table 4. **Ablation of Strategy Components and Metric Choice.** We compare our full strategy against three variants: the Base Detector, a version replacing the GEN score with Shannon Entropy, and a single-threshold version (Ours (w/o τ_{GEN})). The results demonstrate that both the GEN score and our conditional, two-threshold design are critical for performance. Best and second-best results are in **bold** and underlined.

Method	GenImage	Commfor	Avg.
Base Detector	74.55	66.16	70.35
Shannon Entropy	18.05	9.90	13.98
Ours (w/o τ_{GEN})	<u>87.25</u>	<u>94.55</u>	<u>90.90</u>
Ours	95.37	97.49	96.43

detectors built with different VFM backbones, including CLIP-ConvNeXt [22, 35] and DINOv2 [32]. The results in Tab. 6 demonstrate the broad applicability of our approach. Our post-hoc strategy delivers a substantial and consistent average accuracy boost across all architectures, with gains ranging from +15.24% (DINOv2) to +26.08% (CLIP-ViT). This consistency validates our method as a VFM-backbone-agnostic solution, suggesting that the source-asymmetry phenomenon induces the exploitable SACG across diverse VFM architectures.

5. Discussion: On the Generality of SACG

Given that SACG stems from the training data’s source asymmetry, we investigate if this signal exists in other VFM-based detectors. Visualizing their unseen GEN score

Table 5. **Hyperparameter sensitivity analysis.** We evaluate our framework’s robustness by varying three key hyper-parameters: the target $\text{TNR}_{\text{calib}}$, the target TNR_{GEN} , and the **Data Size** (number of seen real images) for calibration. The results demonstrate our method’s stability, highlighting its high data efficiency (optimal with just 128 images) and its insensitivity to the choice of TNR_{GEN} . Best results are in **bold**.

Study	Parameter	GenImage	Commfor	Avg.
$\text{TNR}_{\text{calib}}$	91%	93.73	96.60	95.17
	93%	94.69	97.10	95.89
	95%	95.37	97.49	96.43
	97%	91.98	95.20	93.59
	99%	85.06	85.27	85.17
TNR_{GEN}	1%	95.37	97.49	96.43
	20%	95.37	97.49	96.43
	40%	95.37	97.49	96.43
	60%	95.37	97.49	96.43
Data Size	32	93.42	96.42	94.92
	64	94.73	97.12	95.93
	128	95.37	97.49	96.43
	256	94.73	97.12	95.93
	512	92.35	95.65	94.00

Table 6. **Generality Across VFM Backbones.** We apply our post-hoc strategy (+ Ours) to our base AIGI detectors built from three different VFMs. Our method provides a substantial and consistent performance boost across all architectures, with average accuracy gains from +15.24% to +26.08%. This demonstrates our approach is a VFM-backbone-agnostic solution. Best results are in **bold**.

Vision Models	GenImage	Commfor	Avg.
CLIP-ViT [35]	74.55	66.16	70.35
+ Ours	95.37	97.49	96.43
CLIP-ConvNeXt [22, 35]	77.17	63.24	70.20
+ Ours	93.59	90.68	92.13
DinoV2 [32]	70.92	70.30	70.61
+ Ours	83.76	87.93	85.85

probability densities (Fig. 4) reveals that while all methods exhibit the SACG, its separability varies significantly. Methods with complex training objectives, like Effort [45] (Fig. 4a) and VIBDetect [46] (Fig. 4b), show severe overlap between their real and fake probability densities. We suggest that their sophisticated training targets inadvertently interfere with the SACG. However, a simpler method like Rine [18] (Fig. 4c) exhibits a clean separation similar to our base AIGI detector (Fig. 4d). This provides a key insight: the simplicity of the training schemes is critical, as it preserves the stable, exploitable signal that more complex methods may damage. This also suggests our post-hoc strategy could boost other VFM detectors that are also trained with similar simplicity.

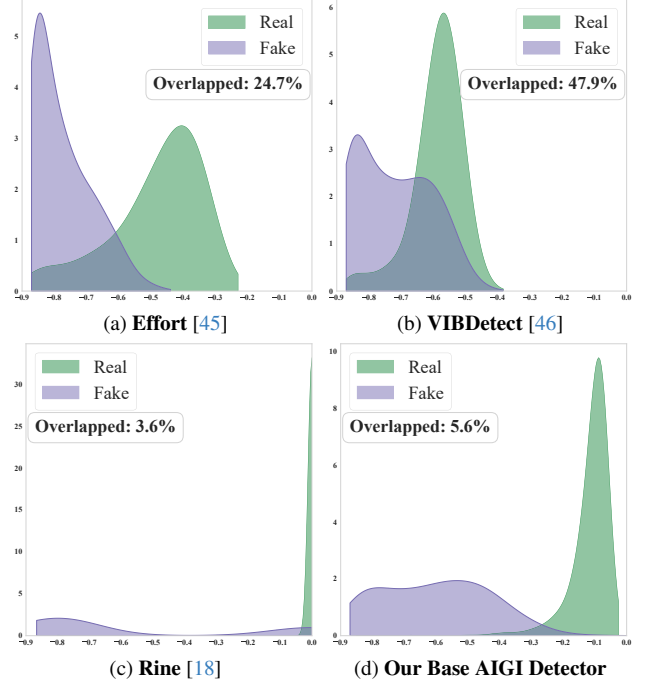


Figure 4. **Probability Density of GEN Scores for Unseen Data.** Probability density plots (y-axis) of the GEN score (x-axis). Complex methods (a) Effort [45] and (b) VIBDetect [46] show severe overlap (24.7% and 47.9%, respectively). Simpler methods (c) Rine [18] and (d) Our Base AIGI Detector show a clean separation (3.6% and 5.6% overlap, respectively). This suggests complex training schemes interfere with the SACG signal.

6. Conclusion

We tackle the AIGI generalization gap by exploiting the *source asymmetry* in the AIGI detection task, demonstrating that this asymmetry induces the **Source-Asymmetry Confidence Gap (SACG)**—a stable separation in predictive confidence between real and fake images. Our primary contribution is a lightweight, post-hoc strategy that efficiently exploits this gap through a simple, real-only calibration. Applied to a partially fine-tuned VFM, this approach decisively outperforms SOTA methods that rely on computationally expensive and complex training workloads. Our work confirms that exploiting this stable signal is a more efficient path to generalization than building increasingly complex detectors. The effectiveness of our post-hoc real-only calibration, however, implicitly assumes that unseen fakes will remain statistically separable from the hard real confidence tail on which our decision rule is calibrated. An advanced generator could potentially exploit this by producing fakes that statistically mimic the confidence distribution of these hard real samples, thus bypassing our detection rule. Investigating the robustness of our calibration against such confidence-mimicking attacks is an important future direction.

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References

- [1] Adobe Inc. Adobe firefly. <https://www.adobe.com/products/firefly>, 2023. 7
- [2] ai-forever. Kandinsky 2.2. <https://github.com/ai-forever/Kandinsky-2>, 2023. 7
- [3] Black Forest Labs. Flux.1. <https://blackforestlabs.ai/>, 2024. 7
- [4] Andrew Brock, Jeff Donahue, and Karen Simonyan. Large scale GAN training for high fidelity natural image synthesis. In *International Conference on Learning Representations*, 2019. 5, 6
- [5] Baoying Chen, Jishen Zeng, Jianquan Yang, and Rui Yang. DRCT: diffusion reconstruction contrastive training towards universal detection of diffusion generated images. In *Proceedings of the International Conference on Machine Learning*, 2024. 2, 5, 6, 7
- [6] Siyuan Cheng, Lingjuan Lyu, Zhenting Wang, Xiangyu Zhang, and Vikash Sehwal. Co-spy: Combining semantic and pixel features to detect synthetic images by ai. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2
- [7] Davide Cozzolino, Giovanni Poggi, Matthias Nießner, and Luisa Verdoliva. Zero-shot detection of ai-generated images. In *European Conference on Computer Vision*, 2024. 3
- [8] Katherine Crowson, Stefan Andreas Baumann, Alex Birch, Tanishq Mathew Abraham, Daniel Z. Kaplan, and Enrico Shippole. Scalable high-resolution pixel-space image synthesis with hourglass diffusion transformers. In *Proceedings of the International Conference on Machine Learning*, 2024. 7
- [9] DeciAI Research Team. Decidiffusion v2.0. <https://huggingface.co/Deci/DeciDiffusion-v2-0>, 2024. 7
- [10] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2009. 5
- [11] Prafulla Dhariwal and Alexander Quinn Nichol. Diffusion models beat gans on image synthesis. In *Advances in Neural Information Processing Systems*, 2021. 5, 6
- [12] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. 3
- [13] Google DeepMind. Imagen. <https://deepmind.google/models/imagen/>, 2024. 5, 6, 7
- [14] Shuyang Gu, Dong Chen, Jianmin Bao, Fang Wen, Bo Zhang, Dongdong Chen, Lu Yuan, and Baining Guo. Vector quantized diffusion model for text-to-image synthesis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. 6
- [15] Fabrizio Guillaro, Giada Zingarini, Ben Usman, Avneesh Sud, Davide Cozzolino, and Luisa Verdoliva. A bias-free training paradigm for more general ai-generated image detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2
- [16] Ideogram, Inc. Ideogram. <https://ideogram.ai/>, 2024. 7
- [17] Dimitrios Karageorgiou, Symeon Papadopoulos, Ioannis Kompatsiaris, and Efstratios Gavves. Any-resolution ai-generated image detection by spectral learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 3
- [18] Christos Koutlis and Symeon Papadopoulos. Leveraging representations from intermediate encoder-blocks for synthetic image detection. In *European Conference on Computer Vision*, 2024. 1, 2, 3, 5, 6, 7, 8
- [19] Kvikontent. Kvikontent midjourney v6 lora. <https://huggingface.co/Kvikontent/midjourney-v6>, 2024. 7
- [20] Bo Liu, Fan Yang, Xiuli Bi, Bin Xiao, Weisheng Li, and Xinbo Gao. Detecting generated images by real images. In *European Conference on Computer Vision*, 2022. 3, 5, 6, 7
- [21] Xixi Liu, Yaroslava Lochman, and Christopher Zach. GEN: pushing the limits of softmax-based out-of-distribution detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. 1, 2, 4, 5
- [22] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. 7, 8
- [23] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. 5
- [24] Simian Luo, Yiqin Tan, Suraj Patil, Daniel Gu, Patrick von Platen, Apolinário Passos, Longbo Huang, Jian Li, and Hang Zhao. Lcm-lora: A universal stable-diffusion acceleration module. arXiv preprint arXiv:2311.05556, 2023. 7
- [25] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 2008. 3
- [26] Midjourney, Inc. Midjourney. <https://www.midjourney.com/>, 2025. 6, 7
- [27] MindSpore. Wukong. <https://xihe.mindspore.cn/modelzoo/wukong/introduce>, 2025. 6
- [28] Alexander Quinn Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. GLIDE: towards photorealistic image generation and editing with text-guided diffusion models. In *Proceedings of the International Conference on Machine Learning*, 2022. 5, 6
- [29] Utkarsh Ojha, Yuheng Li, and Yong Jae Lee. Towards universal fake image detectors that generalize across generative

- models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. 3, 5, 6, 7
- [30] OpenAI. Dall-e 2. <https://openai.com/index/dall-e-2/>, 2022. 7
- [31] OpenAI. Dall-e 3. <https://openai.com/index/dall-e-3/>, 2023. 1, 5, 6, 7
- [32] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mido Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision. *Trans. Mach. Learn. Res.*, 2024. 1, 2, 7, 8
- [33] Jeongsoo Park and Andrew Owens. Community forensics: Using thousands of generators to train fake image detectors. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2, 5, 7
- [34] Pablo Pernias, Dominic Rampas, Mats L. Richter, Christopher Pal, and Marc Aubreville. Würstchen: An efficient architecture for large-scale text-to-image diffusion models. In *International Conference on Learning Representations*, 2024. 7
- [35] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning*, 2021. 1, 2, 3, 5, 7, 8
- [36] Jonas Ricker, Denis Lukovnikov, and Asja Fischer. AER-OBLADE: training-free detection of latent diffusion images using autoencoder reconstruction error. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 3
- [37] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. 1, 5, 6
- [38] Chuangchuang Tan, Huan Liu, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, and Yunchao Wei. Rethinking the up-sampling operations in cnn-based generative network for generalizable deepfake detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 5, 6, 7
- [39] Ming Tao, Hao Tang, Fei Wu, Xiaoyuan Jing, Bing-Kun Bao, and Changsheng Xu. DF-GAN: A simple and effective baseline for text-to-image synthesis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. 7
- [40] Ming Tao, Bing-Kun Bao, Hao Tang, and Changsheng Xu. GALIP: generative adversarial clips for text-to-image synthesis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. 7
- [41] Sheng-Yu Wang, Oliver Wang, Richard Zhang, Andrew Owens, and Alexei A. Efros. Cnn-generated images are surprisingly easy to spot... for now. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020. 2, 5, 6, 7
- [42] Zhendong Wang, Jianmin Bao, Wengang Zhou, Weilun Wang, Hezhen Hu, Hong Chen, and Houqiang Li. DIRE for diffusion-generated image detection. In *IEEE/CVF International Conference on Computer Vision*, 2023. 3, 5, 6, 7
- [43] Qijie Xu, Defang Chen, Jiawei Chen, Siwei Lyu, and Can Wang. Recent advances on generalizable diffusion-generated image detection. arXiv preprint arXiv:2502.19716, 2025. 1
- [44] Shilin Yan, Ouxiang Li, Jiayin Cai, Yanbin Hao, Xiaolong Jiang, Yao Hu, and Weidi Xie. A sanity check for ai-generated image detection. In *International Conference on Learning Representations*, 2025. 2, 5, 6, 7
- [45] Zhiyuan Yan, Jiangming Wang, Peng Jin, Ke-Yue Zhang, Chengchun Liu, Shen Chen, Taiping Yao, Shouhong Ding, Baoyuan Wu, and Li Yuan. Orthogonal subspace decomposition for generalizable ai-generated image detection. In *Proceedings of the International Conference on Machine Learning*, 2025. 1, 2, 3, 5, 6, 7, 8
- [46] Haifeng Zhang, Qinghui He, Xiuli Bi, Weisheng Li, Bo Liu, and Bin Xiao. Towards universal ai-generated image detection by variational information bottleneck network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 1, 2, 5, 6, 7, 8
- [47] Mingjian Zhu, Hanting Chen, Qiangyu Yan, Xudong Huang, Guanyu Lin, Wei Li, Zhijun Tu, Hailin Hu, Jie Hu, and Yunhe Wang. Genimage: A million-scale benchmark for detecting ai-generated image. In *Advances in Neural Information Processing Systems*, 2023. 2, 5, 6